

# 1.10: Differential Equations

## 1. Terminology and Classification

A differential equation (DE) relates a function to its derivatives. The **order** is the highest derivative. A DE is **linear** if the unknown and its derivatives appear only to the first power and not multiplied together. Linear DEs are **homogeneous** if the RHS is zero.

OCR B (MEI) covers first-order linear DEs, second-order linear DEs with constant coefficients, and coupled first-order systems.

## 2. First-Order Linear DEs — Integrating Factor

For a DE of the form

$$\frac{dy}{dx} + P(x)y = Q(x),$$

multiply through by the **integrating factor**

$$\mu(x) = e^{\int P(x) dx}.$$

The left side becomes  $\frac{d}{dx}(\mu y)$ , so

$$\mu y = \int \mu(x) Q(x) dx + C.$$

**Solve**  $\frac{dy}{dx} + \frac{2y}{x} = x$ .

Here  $P = 2/x$ , so  $\mu = e^{\int 2/x dx} = e^{2 \ln x} = x^2$ .

Multiply:  $x^2 y' + 2xy = x^3 \Rightarrow \frac{d}{dx}(x^2 y) = x^3$ .

Integrate:  $x^2 y = \frac{x^4}{4} + C \Rightarrow y = \frac{x^2}{4} + \frac{C}{x^2}$ .

### 3. Second-Order Linear Homogeneous DEs with Constant Coefficients

For

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0,$$

try  $y = e^{mx}$  to obtain the **auxiliary equation**  $am^2 + bm + c = 0$ .

| Roots                        | General solution                    |
|------------------------------|-------------------------------------|
| Two distinct real $m_1, m_2$ | $y = Ae^{m_1 x} + Be^{m_2 x}$       |
| Repeated real $m$            | $y = (A + Bx)e^{mx}$                |
| Complex $m = p \pm iq$       | $y = e^{px}(A \cos qx + B \sin qx)$ |

**Solve**  $y'' - 4y' + 13y = 0$ .

Auxiliary:  $m^2 - 4m + 13 = 0 \Rightarrow m = 2 \pm 3i$ .

GS:  $y = e^{2x}(A \cos 3x + B \sin 3x)$ .

### 4. Non-Homogeneous Second-Order DEs

For  $ay'' + by' + cy = f(x)$  the general solution is

$$y = y_{CF} + y_{PI}$$

- $y_{CF}$  (complementary function): the GS of the *homogeneous* version (§3).
- $y_{PI}$  (particular integral): any single function satisfying the full DE.

## Trial Functions for PI

| $f(x)$                             | Trial $y_{PI}$                                                       |
|------------------------------------|----------------------------------------------------------------------|
| Polynomial of degree $n$           | General polynomial of degree $n$                                     |
| $e^{\lambda x}$                    | $Pe^{\lambda x}$ (unless $\lambda$ is already a root of the aux eqn) |
| $\sin \omega x$ or $\cos \omega x$ | $P \cos \omega x + Q \sin \omega x$                                  |
| Combination                        | Sum of the corresponding trials                                      |

If the chosen trial form is already part of  $y_{CF}$ , multiply by  $x$  (or  $x^2$  for a repeated root) before substituting.

**Solve**  $y'' - 3y' + 2y = 4e^{3x}$ .

CF:  $m^2 - 3m + 2 = 0 \Rightarrow m = 1, 2$ .  $y_{CF} = Ae^x + Be^{2x}$ .

PI: try  $y = Pe^{3x}$ .  $y' = 3Pe^{3x}$ ,  $y'' = 9Pe^{3x}$ .

Substitute:  $(9P - 9P + 2P)e^{3x} = 4e^{3x} \Rightarrow 2P = 4 \Rightarrow P = 2$ .

GS:  $y = Ae^x + Be^{2x} + 2e^{3x}$ .

## Boundary/Initial Conditions

Use to pin down  $A, B$ . Apply after writing the GS.

## 5. Coupled First-Order Linear Systems

A pair like

$$\frac{dx}{dt} = 3x + y, \quad \frac{dy}{dt} = -2x + y$$

can be solved by **elimination**: differentiate one equation and substitute to get a second-order DE in a single variable.

**From the system above:**

Differentiate the first:  $x'' = 3x' + y'$ . Substitute  $y' = -2x + y$  and  $y = x' - 3x$ :

$$x'' = 3x' + (-2x) + (x' - 3x) = 4x' - 5x.$$

So  $x'' - 4x' + 5x = 0$ . Aux:  $m^2 - 4m + 5 = 0 \Rightarrow m = 2 \pm i$ .

$$x = e^{2t}(A \cos t + B \sin t), \text{ and } y = x' - 3x.$$

Alternatively (richer option — FP2 level): write as a matrix system  $\dot{\mathbf{z}} = \mathbf{A}\mathbf{z}$  and diagonalise (see §1.3).

## 6. Modelling with Differential Equations

### Newton's Law of Cooling

$$\frac{dT}{dt} = -k(T - T_{\text{env}})$$

Separable:  $T = T_{\text{env}} + Ce^{-kt}$ .

### Exponential Growth/Decay

$$\frac{dN}{dt} = kN \Rightarrow N = N_0 e^{kt}$$

### Mechanical Oscillators (Damped Harmonic Motion)

$$m\ddot{x} + c\dot{x} + kx = F(t)$$

- **Underdamped** ( $c^2 < 4mk$ ): complex roots — oscillation with decaying amplitude.
- **Critically damped** ( $c^2 = 4mk$ ): repeated real root — fastest non-oscillating decay.
- **Overdamped** ( $c^2 > 4mk$ ): two real negative roots — slow decay without oscillation.

### Resonance

When the driving frequency matches the natural frequency of an undamped oscillator, the PI trial  $P \cos \omega t + Q \sin \omega t$  fails because it's already in  $y_{CF}$ . Multiplying by  $t$  gives unbounded growth — *resonance*.

## 7. Exam Style Strategy

- First-order linear: identify  $P(x)$ , compute  $\mu$ , *don't forget* the constant of integration.
- Second-order: always write the auxiliary equation and classify its roots before the PI trial.
- If your PI trial clashes with the CF, multiply by  $x$  — this is the single most common mistake.
- Always apply initial conditions *after* writing the GS, and simplify.
- In modelling questions, interpret constants physically (e.g. " $k > 0$  represents the cooling rate").